

INFO 5368-030: Practical Applications in Machine Learning



CORNELL TECH

Due: May 12th at 11PM

Assignment Type: Group (Up to 6 members)

Submit on report: Gradescope (Entry code: 7KB3DP); Submit PDF.

Submit Code: <https://classroom.github.com/a/dJBAHCPL>

Total Points: 137 points

NO LATE SUBMISSIONS ALLOWED

Final Report Report, Code, & Deployed Application

Overview

In final projects, students build a machine learning application that addresses a real-world problem. Students select a well-justified problem domain (e.g., housing, healthcare, social media), identify a publicly available dataset for training and testing a machine learning model, identify a machine learning model to solve the problem, and standard evaluate metrics to evaluate model performance. Then, students build a Streamlit application that deploys the trained model. The **objective** of final projects include:

- Choose a problem to address using machine learning.
- Identify and justify the machine learning algorithms to solve the problem.
- 2 or more machine learning algorithms for comparison.
- Identify evaluation metrics to measure the performance of the algorithms.
- Conduct experiments using the machine learning models and evaluation metrics to identify the best performing algorithm.
- Build an online web application using Streamlit that deploys the best performing model.

Quick Links

- [Overleaf template](#)
- [Final Project Streamlit Templates](#)

Report Outline

- Title (1 point)
- Abstract (5 points)
- Introduction (9 points)
- Background (5 points)
- End-to-End ML Pipeline (41 points)
- Team Member Contribution (5 points)
- Code, Documentation, and App Deployment (60 points)

Final Project Report Outline

0. Title (1 point)

Include the teams members' names, program, university affiliation; and a project title that effectively describes the project goals.

1. Abstract (5 points)

- **Introduction/Background:** Briefly introduce the topic and its significance, highlighting the problem or question your research addresses. (1 point)
- **Purpose/Research Question:** State the main objective or research question of your project. (1 point)
- **Methods:** Briefly describe two or more machine learning methods and three or more metrics used in your project. (1 point)
- **Summarize Results:** Discuss key findings of your project, focusing on the most important outcomes. (1 point)
- **Significance:** Discuss the implications or impact of your project, emphasizing its relevance and contribution to the field. (1 point)

2. Introduction (9 points)

Establish the problem or issue you want to research

- Highlight the importance of the problem/issue to solve (Why should we care about this problem?). (1 point)
- Present an overview on current research on the issue or problem. (1 point)

Summarize prior efforts to solve the problem

- Present an overview on current research on the issue or problem. (1 point)
- Identify a gap, problems or unresolved issues in the existing knowledge/research that your project fills or identify a useful technical focus. (1 point)
- Cite relevant literature using bibtex (e.g., on Google Scholar). Enter the bibtex code in the references.bib file in Overleaf and cite the source (e.g., `\cite{example_bibtex_label}`). (1 point)

Summarize the ML approach

- State and justify the chosen solution including 2 or more machine learning methods. (1 point)
- Indicate 3 or more evaluation metrics to measure success of solving the problem of interest. (1 point)
- Provide details about the developed Streamlit application and its intended use by a target user group. (1 point)

Broader Impact of the project

- What will the machine learning application contribute to the field, society, and beyond? (1 point)

3. Background (5 points)

- **Project Scope:** Present a general overview of the problem to solve, its significance, and main aims; this may be similar to establishing the “importance of the topic” in the introduction. (1 point)
- **Comparisons to prior work:** Discuss the existing level of research done on the research topic or on related topics in the field to set context for your project. Be concise and mention only the relevant part of studies, ideally in chronological order to reflect the progress being made. (1 point)
- **Share common knowledge:** Highlight disputes in the field as well as claims made by scientists, organizations, or key stakeholders that need to be investigated. This forms the foundation of your methodology and solidifies the aims of your project. (1 point)
- **Identify knowledge gaps:** Describe if and how the methods and techniques used in the research study are different from those used in previous research on similar topics. Based on prior work, describe how your approach compares to prior approaches. (1 point).
- **Cite relevant literature** using bibtex (e.g., on Google Scholar). Enter the bibtex code in the references.bib file in Overleaf and cite the source (e.g., `\cite{example_bibtex_label}`). (1 point)

4. End-to-End ML Pipeline

4.1 Offline Model Training and Evaluation

Project reports must describe offline development of machine learning algorithms to solve the project goals.

4.1.1 Data Collection, Exploration & Processing (12 points)

- State the name of the dataset(s) used to train the machine learning models including
 - Data types (e.g., numerical, categorical, image, text, and audio data) (1 point)
 - Quantity of data (1 point)
 - Year it was released (if available) (1 point)
 - Source of the dataset (e.g., Kaggle) (1 point)
 - How the data was used in the project. (1 point)
- State whether ground truth labels are provided and what data is used as ground truth. If there are none, state how you generated ground truth data or provide justification for the evaluation procedure. At least one dataset is required. (1 point)
- State and justify **data exploration** methods used to gain knowledge about the dataset including figures/plots such as Scatterplots, Histograms, Boxplots, Lineplots, or others. Include figures to justify these insights. (3 points)
- State and justify **preprocessing** techniques used to prepare the dataset for ML algorithms including data cleaning (e.g., handling missing data), feature encoding (one-hot, integer),

feature normalization, correlations, outlier removal, dimensionality reduction, feature creation, discretization, binarization, and augmentation. (3 points)

4.1.2 Methods and Model Training (9 points)

- Describe 2 or more machine learning algorithms used to solve the problem of interest. Options of machine learning algorithms include regression, classification, clustering, and artificial neural networks (**NO exceptions**). **Use of machine learning libraries for training models is prohibited (NO exceptions)**. (3 point)
- Discuss the problem formulation, relevant equations, and theoretical underpinnings. (3 points)
- Justify why the machine learning algorithms are well-suited to solve the problem. (1 point)
- Describe model inputs and outputs. (2 points)

4.1.3 Model Evaluation (7 points)

- Describe at least **three evaluation metrics** used to measure the success of solving the problem including relevant equations. (3 points)
- Provide justification for why the evaluation metrics are well-suited to measure success. (1 point)
- Describe the training and validation procedure including the train/test split. Make sure that the experimental setup is described in enough detail to reproduce your results. (1 point)
- Describe the experiments including fine-tuning model hyperparameters. (1 point)
- Discuss the methods used to avoid model overfitting and underfitting. (1 point)

4.1.4 Results (10 points)

Summarize the experimental results including the best and worst performing algorithms for at least 3 evaluation metrics. (10 points)

4.1.5 Model Deployment & Front-End Application Using Streamlit (6 points)

- Describe the web-based application you developed using the Streamlit library. (1 point)
- Describe and justify how the deployed model will be chosen. The machine learning algorithms may have tradeoffs in performance in terms of the selected evaluation metrics. Provide details about how the model was selected given these trade-offs. (1 point)
- Describe the target population and how the application built is beneficial to them. (1 point)

- Describe the user interface (UI) layout including user inputs (using menus, buttons, etc) and interface outputs. (1 point)
- Provide figures or screenshots to show the website layout and provide detailed discussion on how the front- and back-end are connected (e.g., deploy a model by pressing a button). (2 points)

4.1.6 Conclusion (5 points)

- Summarize the project goals, ML approach, and key experimental results. (3 points)
- Provide discussion of the implications of project findings for your target user population and society broadly. (1 point)
- State future plans for this project, applying lessons learned from model training and evaluation. (1 point)

7. Team Member Contribution (5 points)

Each team member is expected to contribute to the final project. Task responsibilities must be clearly stated in terms of offline model training and evaluation, and front-end development.

Technical Components: Include the team contributions in terms of technical goals of the project such as programming offline, front- and back-end components and algorithms.

Writing Components: Include the team contributions for writing sections of the final report for all sections.

8. Code, Documentation, and Deployment (60 points)

The final report should include a link to a GitHub repository with your code. If you prefer to keep your code private, submit a Google Drive URL. Your code should include the following:

- Jupyter notebook with your experiments described in the report. The teaching team will run your code to ensure the results are consistent with the results reported. (20 points)
- Streamlit application code that is consistent with the discussion in the final project report. (15 points)
- Dataset used to train and evaluate the machine learning models. (10 points)
- README that documents the repositories contents and with instructions to run the Streamlit application. (5 points)
- Submit URL to a successfully deployed web application with no errors. Store your code and launched application on the final report [GitHub Classroom](#). (10 points)

9. Report Writing, Formatting, and Figures

9.1 Writing Clarity

Written reports will be evaluated based on clarity, significance of the problem, and the technical quality of the work (the level of depth in the experimental analyses, does the approach make sense to solve the problem, are the algorithms explained in enough detail, etc).

Remember this is a formal document. It should have correct grammar and should be spell checked. It should have figures that are clear, easy to read, and are referenced in the text.

It should adopt a formal writing tone - avoid slang, use active voice, use plural first person (“we” or “our team”), clear, and concise.

Use complete sentences throughout the document as if this content is expected to be published to a public audience. Recall the writing rule for paragraph structure (you probably learned in grade school): One main idea + three supporting sentences. If it gets beyond that, it's time for a new paragraph.

Lastly, make sure to define acronyms before using them.

Submit the report via Gradescope and make sure to submit as a team.

If you would like writing help, please try [these resources at Cornell](#).

9.2 Figures/Tables

Include and label all your figures and tables and include captions to describe them. This should include: title, axes, units, legend (if there is more than one plot in the figure). Refer to all figures in the text so readers know why the figures were included in the document.

9.3 Format

The project reports require the following format using this IEEE [Overleaf template](#):

1. Double column
2. Single-spaced
3. 1-inch margin
4. 12 pt font size
5. Up to 8 pages (excluding references)